

Magnetic Field in Matter

What is magnetism?

General answer: horse shoe magnet, compass needle, north pole



Loadstone

1

Magnetic Field in Matter

What is magnetism?

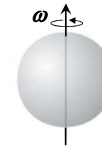
General answer: horse shoe magnet, compass needle, north pole

What is the connection of these with moving charges/current Carrying wires?

Obviously nothing.....at least macroscopically!

But all magnetic phenomenon are due to charges in motion!

What's going on? How about microscopically?



2

Magnetic Field in Matter

What is magnetism?

General answer: horse shoe magnet, compass needle, north pole

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Obviously nothing.....at least macroscopically!

But all magnetic phenomenon are due to charges in motion!

What's going on? How about microscopically?

Electrons orbiting around nuclei, and spinning about their axis comprising tiny current loops

macroscopically can be treated as magnetic dipole !



3

Magnetic Field in Matter

In general these dipoles are randomly oriented, and cancel each other.

With the application of the external magnetic field, individual dipole moments get aligned in the direction of the field

Though the alignment not complete.....medium becomes magnetically polarized or **MAGNETIZED**

This phenomenon is called **MAGNETIZATION**

Magnetization parallel to **B** (Para-magnetization)

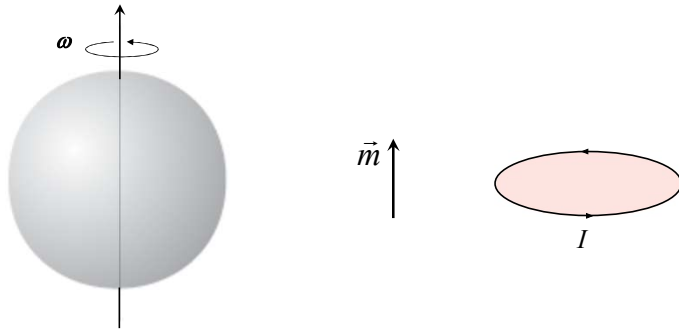
Magnetization anti parallel to **B** (Dia-magnetization)

Retains magnetization even after the removal of **B**
(Ferro-magnetization)

4

Magnetic Field in Matter

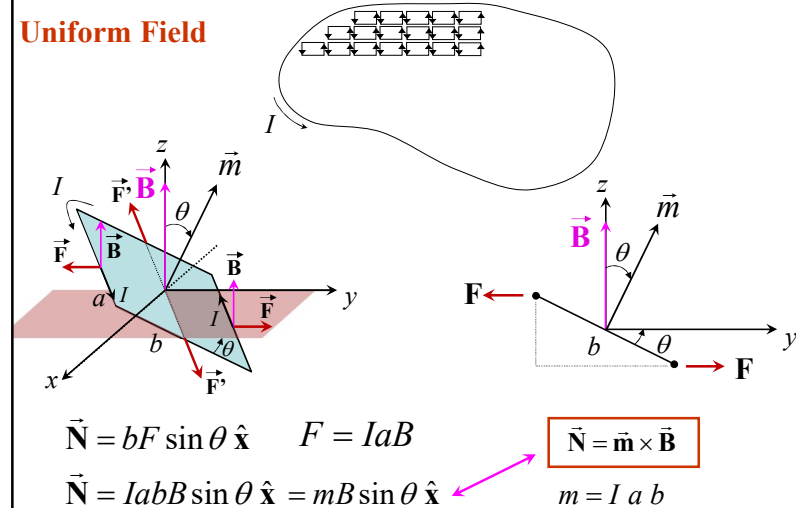
Effect of spinning electron in **Uniform Field**



5

Torque and force on magnetic dipole

Uniform Field



6

Torque and force on magnetic dipole

Uniform Field

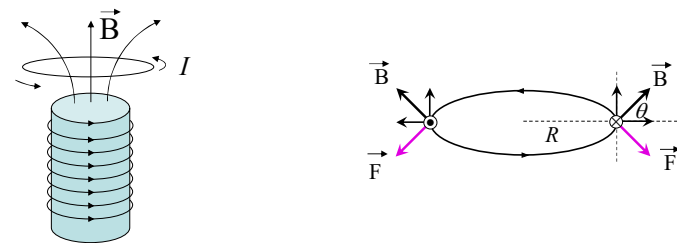
$$\vec{N} = \vec{m} \times \vec{B}$$

- Force on the dipole: $\vec{F} = I \oint d\vec{l} \times \vec{B} = I(\oint d\vec{l}) \times \vec{B} = \vec{0}$
- Tends to align the dipole in the direction of external field
- **Responsible for Para-magnetism**
- Every **spinning** electron in an atom constitute a magnetic dipole
- Is para-magnetism a universal phenomenon ?
- No. Essentially, you must have an atom/molecule with odd number of electrons

7

Torque and force on magnetic dipole

Non-Uniform Field



Net downward force: $|\vec{F}| = \left| I \oint d\vec{l} \times \vec{B} \right| = I \oint dl B \cos \theta \sin 90^\circ$
 $= I \oint dl B \cos \theta = 2\pi RIB \cos \theta$

$$\vec{F} = \nabla(\vec{m} \cdot \vec{B})$$

8

Torque and force on magnetic dipole

Uniform Field

$$\vec{N} = \vec{m} \times \vec{B}$$

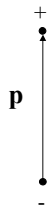
$$\vec{F} = 0$$

Non-Uniform Field

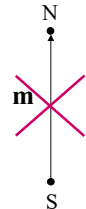
$$\vec{N} = \vec{m} \times \vec{B}$$

Exact torque about the center for a perfect dipole

$$\vec{F} = \vec{\nabla}(\vec{m} \cdot \vec{B})$$



electric dipole



magnetic counterpart

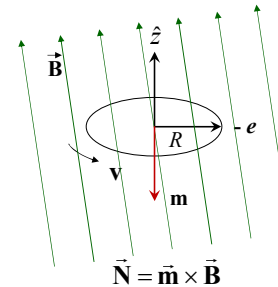


real magnetic dipole

9

Effect of magnetic Field on Orbital motion

We analyzed effect of spinning electron. But electron also revolve around the nucleus → Orbital Motion



$$T = \frac{2\pi R}{v}$$

$$I = \frac{e}{T} = \frac{ev}{2\pi R}$$

$$\vec{m} (= I \pi R^2) = -\frac{1}{2} e v R \hat{z}$$

Tends to align $\vec{m}$ along $\vec{B}$ → possible only by tilting entire orbit !

Orbital contribution to Paramagnetism is extremely small.

10

Effect of magnetic Field on Orbital motion

Lets take a deeper look of the effect of Orbital motion

Centripetal force $m_e \frac{v^2}{R} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2}$

Consider $\vec{B}$ normal to the plane of the orbit !

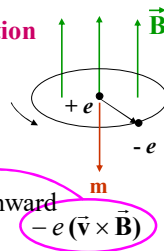
$$m_e \frac{\bar{v}^2}{R} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2} + e \bar{v} B \quad \text{inward}$$

$$e \bar{v} B = m_e \frac{\bar{v}^2}{R} - \frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2} = m_e \frac{\bar{v}^2}{R} - m_e \frac{v^2}{R}$$

$$= \frac{m_e}{R} (\bar{v}^2 - v^2) = \frac{m_e}{R} (\bar{v} - v)(\bar{v} + v) = \frac{m_e}{R} (\Delta v) 2\bar{v}$$

Change in the Velocity: $\Delta v = \frac{eRB}{2m_e}$

With $\vec{B}$ on, electron speeds up!



11

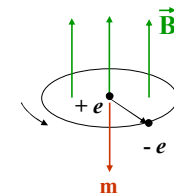
Effect of magnetic Field on Orbital motion

Change in the Velocity: $\Delta v = \frac{eRB}{2m_e}$

With $\vec{B}$ on, electron speeds up!

$$\Delta \vec{m} = -\frac{1}{2} e (\Delta v) R \hat{z} = -\frac{e^2 R^2}{4m_e} \vec{B}$$

Change in $\vec{m}$ is opposite to the direction of $\vec{B}$!



12

Effect of magnetic Field on Orbital motion

$$\vec{m} (= I \pi R^2 \hat{z}) = +\frac{1}{2} e v R \hat{z}$$

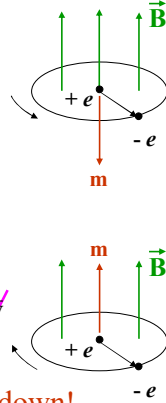
$$m_e \frac{\bar{v}^2}{R} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2} - e \bar{v} B \quad -e(\bar{v} \times \vec{B}) \text{ outward}$$

$$e \bar{v} B = \frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2} - m_e \frac{\bar{v}^2}{R} = m_e \frac{v^2}{R} - m_e \frac{\bar{v}^2}{R}$$

$$= \frac{m_e}{R} (v^2 - \bar{v}^2) = \frac{m_e}{R} (v - \bar{v})(v + \bar{v}) = -\frac{m_e}{R} (\Delta v) 2v$$

$$\Delta v = -\frac{eRB}{2m_e} \quad \text{With } \vec{B} \text{ on, electron slowed down!}$$

$$\Delta \vec{m} = \frac{1}{2} e (\Delta v) R \hat{z} = -\frac{e^2 R^2}{4m_e} \vec{B} \quad \text{Still, change in } \vec{m} \text{ is opposite to the direction of } \vec{B} !$$



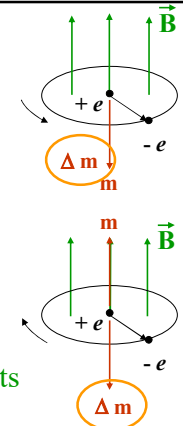
13

Effect of magnetic Field on Orbital motion

- With $\vec{B}$ on, electron speeds up!
- Change in $\vec{m}$ is opposite to the direction of $\vec{B}$!
- With $\vec{B}$ on, electron slowed down!
- Still, change in $\vec{m}$ is opposite to the direction of $\vec{B}$!

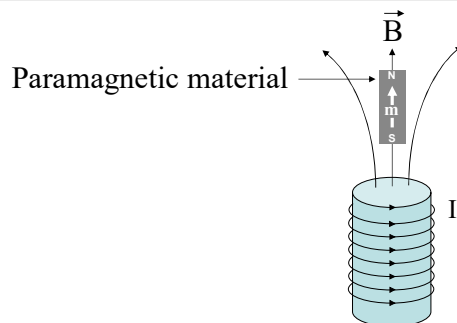
In general, electron orbits are randomly oriented, canceling individual dipole moments

With $\vec{B}$ on, each atom picks up little “extra” dipole moment increment anti-parallel to $\vec{B}$ Responsible for Diamagnetism



14

Magnetic material in external field



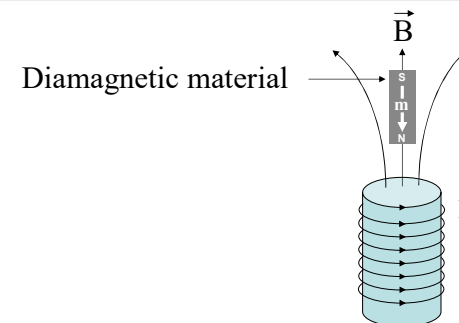
Induced dipole moment → upward

Force on the object → downward

Paramagnet is attracted into the field

15

Magnetic material in external field



Induced dipole moment → downward

Force on the object → upward

Diamagnet is repelled away by the field

16

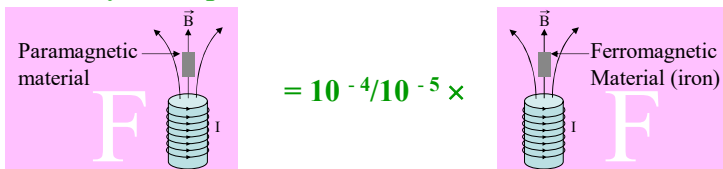
Magnetic material in external field

Strength of diamagnetism (alteration of orbital speed in such a way to give rise the change in orbital dipole moment in the direction opposite to the external field)

much weaker than

Strength of paramagnetism (torque on dipole associated with the spin of unpaired electron to align dipole parallel to the external field)

In general, diamagnetism and paramagnetism are extremely weak phenomenon



17

Magnetic material in external field - Conclusion

With the application of the external magnetic field, medium becomes magnetically polarized or **MAGNETIZED**

This phenomenon is called **MAGNETIZATION**

Magnetization: $\vec{M}$ = magnetic dipole per unit volume

18

Field of Magnetized Object

Magnetized material

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$$

$$d\vec{m}(\vec{r}') = \vec{M}(\vec{r}') d\tau'$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{M}(\vec{r}') \times \hat{r}}{r^2} d\tau'$$

$$r = [(x-x')^2 + (y-y')^2 + (z-z')^2]^{1/2}$$

$$\vec{\nabla}' \left(\frac{1}{r} \right) = \frac{\hat{r}}{r^2}$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \vec{M}(\vec{r}') \times \vec{\nabla}' \left(\frac{1}{r} \right) d\tau'$$

19

Field of Magnetized Object

Magnetized material

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \vec{M}(\vec{r}') \times \vec{\nabla}' \left(\frac{1}{r} \right) d\tau'$$

$$\vec{\nabla} \times (f\vec{A}) = f(\vec{\nabla} \times \vec{A}) - \vec{A} \times (\vec{\nabla} f)$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \left[\int \frac{1}{r} [\vec{\nabla}' \times \vec{M}(\vec{r}')] d\tau' - \int \vec{\nabla}' \times \left(\frac{\vec{M}(\vec{r}')}{r} \right) d\tau' \right]$$

$$\int_V [\vec{\nabla} \cdot \vec{V}] d\tau = \oint_S \vec{V} \cdot d\vec{a}$$

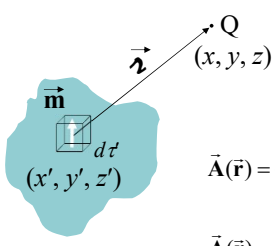
$$\int_V [\vec{\nabla} \cdot (\vec{c} \times \vec{V})] d\tau = \oint_S (\vec{c} \times \vec{V}) \cdot d\vec{a}$$

$$\int_V [\vec{V} \cdot (\vec{\nabla} \times \vec{c}) - \vec{c} \cdot (\vec{\nabla} \times \vec{V})] d\tau = \oint_S d\vec{a} \cdot (\vec{c} \times \vec{V})$$

$$- \int_V \vec{c} \cdot (\vec{\nabla} \times \vec{V}) d\tau = \oint_S \vec{c} \cdot (\vec{V} \times d\vec{a}) \rightarrow - \int_V (\vec{\nabla} \times \vec{V}) d\tau = \oint_S (\vec{V} \times d\vec{a})$$

20

Field of Magnetized Object



Magnetized material

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \vec{M}(\vec{r}') \times \vec{\nabla}' \left(\frac{1}{r} \right) d\tau'$$

$$\vec{\nabla} \times (f\vec{A}) = f(\vec{\nabla} \times \vec{A}) - \vec{A} \times (\vec{\nabla} f)$$

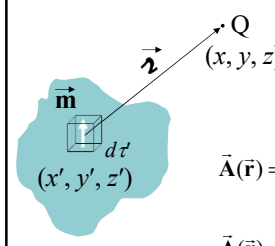
$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \left[\int \frac{1}{r} [\vec{\nabla}' \times \vec{M}(\vec{r}')] d\tau' - \int \vec{\nabla}' \times \left(\frac{\vec{M}(\vec{r}')}{r} \right) d\tau' \right]$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{1}{r} [\vec{\nabla}' \times \vec{M}(\vec{r}')] d\tau' + \frac{\mu_0}{4\pi} \oint_S \frac{1}{r} [\vec{M}(\vec{r}') \times d\vec{a}]$$

$$-\int_V \vec{\nabla} \cdot (\vec{\nabla} \times \vec{V}) d\tau = \oint_S (\vec{\nabla} \times \vec{V}) \cdot d\vec{a} \rightarrow -\int_V (\vec{\nabla} \times \vec{V}) d\tau = \oint_S (\vec{V} \times d\vec{a})$$

21

Field of Magnetized Object



Magnetized material

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \vec{M}(\vec{r}') \times \vec{\nabla}' \left(\frac{1}{r} \right) d\tau'$$

$$\vec{\nabla} \times (f\vec{A}) = f(\vec{\nabla} \times \vec{A}) - \vec{A} \times (\vec{\nabla} f)$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \left[\int \frac{1}{r} [\vec{\nabla}' \times \vec{M}(\vec{r}')] d\tau' - \int \vec{\nabla}' \times \left(\frac{\vec{M}(\vec{r}')}{r} \right) d\tau' \right]$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{1}{r} [\vec{\nabla}' \times \vec{M}(\vec{r}')] d\tau' + \frac{\mu_0}{4\pi} \oint_S \frac{1}{r} [\vec{M}(\vec{r}') \times d\vec{a}]$$

Potential due to volume current

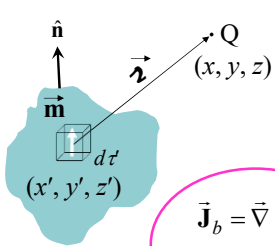
Potential due to surface current

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}')}{r} d\tau'$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \oint \frac{\vec{K}}{r} da'$$

22

Field of Magnetized Object



Magnetized material

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{1}{r} [\vec{\nabla}' \times \vec{M}(\vec{r}')] d\tau' + \frac{\mu_0}{4\pi} \oint_S \frac{1}{r} [\vec{M}(\vec{r}') \times d\vec{a}]$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}_b(\vec{r}')}{r} d\tau' + \frac{\mu_0}{4\pi} \oint_S \frac{\vec{K}_b(\vec{r}')}{r} da'$$

$\vec{J}_b = \vec{\nabla} \times \vec{M}$ Bound volume current density

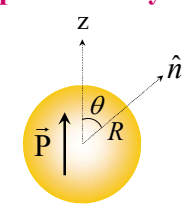
$\vec{K}_b = \vec{M} \times \hat{n}$ Bound surface current density

$\rho_b = -\vec{\nabla} \cdot \vec{P}$ $\sigma_b = \vec{P} \cdot \hat{n}$

23

Example from electrostatics (Example 4.2) to recall

$\vec{E}$ produced by a uniformly polarized sphere of radius R



$r < R$

$$\vec{E} = -\vec{\nabla} V = -\frac{P}{3\epsilon_0} \hat{z} = -\frac{1}{3\epsilon_0} \vec{P}$$

Field is constant inside and $\uparrow \downarrow \vec{P}$

$r > R$

$$V = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$$

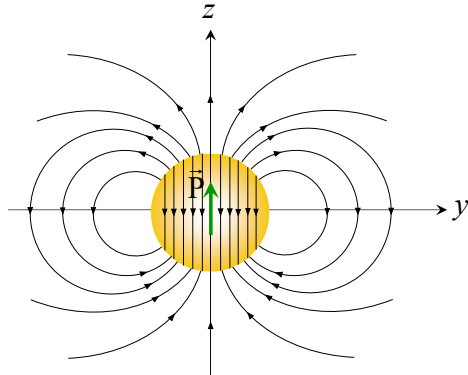
Potential of a perfect dipole at the origin

$$\vec{p} = \frac{4}{3} \pi R^3 \vec{P}$$

24

Example from electrostatics (Example 4.2) to recall

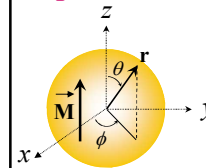
$\vec{E}$ produced by a uniformly polarized sphere of radius R



25

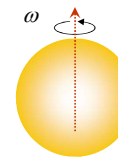
Example for magnetostatics

$\vec{B}$ produced by a uniformly magnetized sphere of radius R



$$\vec{J}_b = \vec{\nabla} \times \vec{M} = 0$$

$$\vec{K}_b = \vec{M} \times \hat{n} = M \sin \theta \hat{\phi}$$



A spinning spherical shell of uniform surface density σ

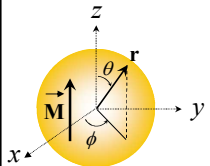
→ Surface current density $\vec{K} = \sigma \vec{v} = \sigma R \omega \sin \theta \hat{\phi}$
 $\sigma R \vec{\omega} \rightarrow \vec{M}$

Field of uniformly magnetized sphere is identical to the field of a spinning spherical shell

26

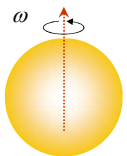
Example for magnetostatics

$\vec{B}$ produced by a uniformly magnetized sphere of radius R



$$\vec{J}_b = \vec{\nabla} \times \vec{M} = 0$$

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A spinning spherical shell of uniform surface density σ

→ Surface current density $\vec{K} = \sigma \vec{v} = \sigma R \omega \sin \theta \hat{\phi}$
 $\sigma R \vec{\omega} \rightarrow \vec{M}$

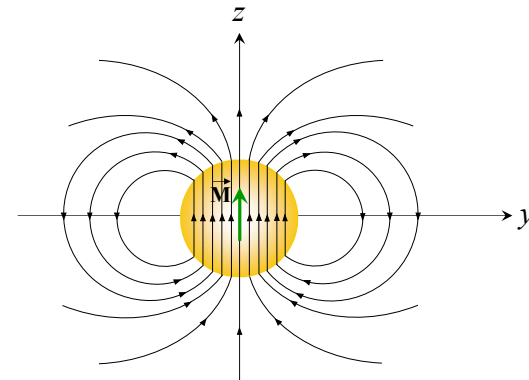
inside $\vec{B} = \frac{2}{3} \mu_0 \sigma R \vec{\omega} = \frac{2}{3} \mu_0 \vec{M}$ Uniform field [Ex:5.11]
 field of a spinning spherical shell

outside $\vec{m} = \frac{4}{3} \pi \sigma R \vec{\omega} R^3 = \frac{4}{3} \pi R^3 \vec{M}$ pure dipole [HW]

27

Example for magnetostatics

$\vec{B}$ produced by a uniformly magnetized sphere of radius R



28

$\vec{H}$

$$\vec{J} = \vec{J}_b + \vec{J}_f \quad \vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

$$\frac{1}{\mu_0} (\vec{\nabla} \times \vec{B}) = \vec{J} = \vec{J}_b + \vec{J}_f = \vec{J}_f + \vec{\nabla} \times \vec{M}$$

$$\vec{\nabla} \times \left(\frac{1}{\mu_0} \vec{B} - \vec{M} \right) = \vec{J}_f$$

$$\vec{H} = \left(\frac{1}{\mu_0} \vec{B} - \vec{M} \right)$$

$$\vec{\nabla} \times \vec{H} = \vec{J}_f$$

$$\oint \vec{H} \cdot d\vec{l} = I_{f_{enc}}$$

Ampere's Law for $\vec{H}$

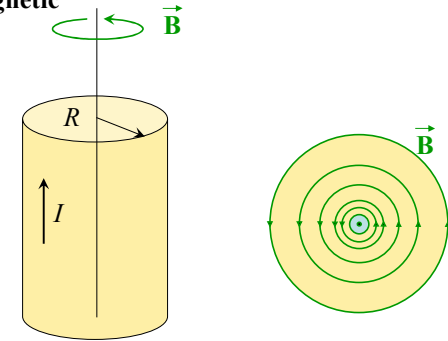
Apply if $\vec{B}$ or $\vec{H}$ needed when there is high degree of symmetry

29

Example

Long copper rod with uniformly distributed (free) current I
 $\vec{H}$ (inside and outside)?

Copper is weakly diamagnetic

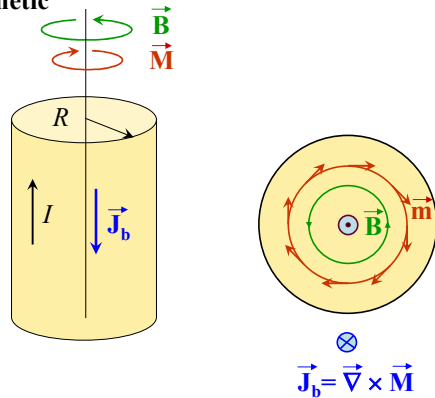


30

Example

Long copper rod with uniformly distributed (free) current I
 $\vec{H}$ (inside and outside)?

Copper is weakly diamagnetic

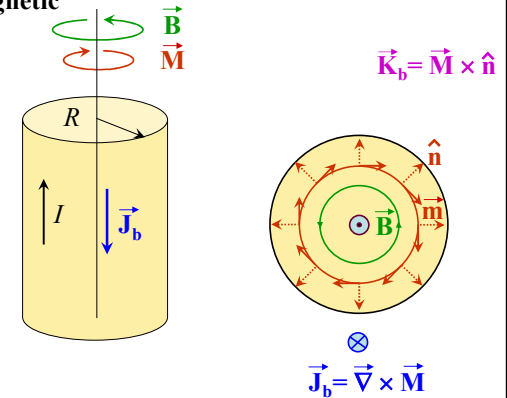


31

Example

Long copper rod with uniformly distributed (free) current I
 $\vec{H}$ (inside and outside)?

Copper is weakly diamagnetic

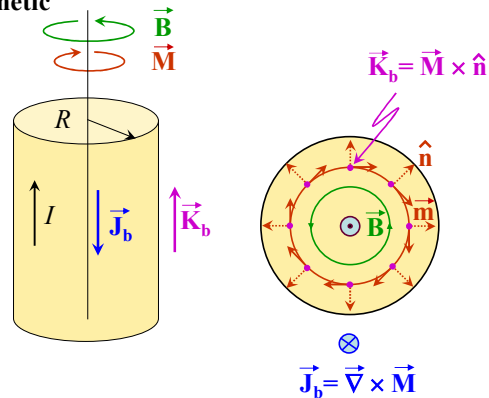


32

Example

Long copper rod with uniformly distributed (free) current I $\vec{H}$ (inside and outside)?

Copper is weakly diamagnetic



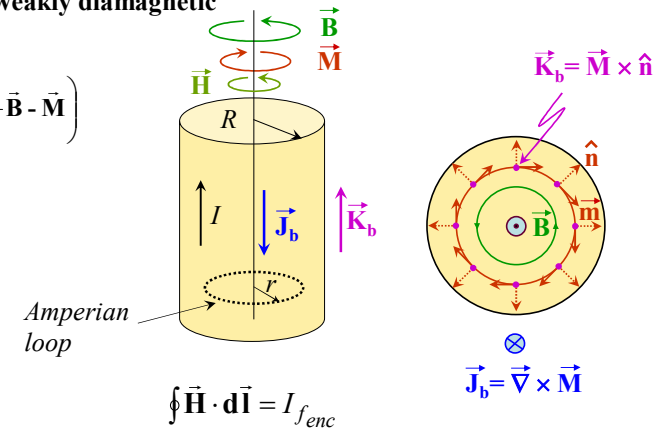
33

Example

Long copper rod with uniformly distributed (free) current I $\vec{H}$ (inside and outside)?

Copper is weakly diamagnetic

$$\vec{H} = \left(\frac{1}{\mu_0} \vec{B} - \vec{M} \right)$$



H. E.

$$\oint \vec{H} \cdot d\vec{l} = I_{fenc}$$

34

Constituting Relation

$$\vec{M} = \chi_m \vec{H}$$

→ **Magnetic Susceptibility** (dimensionless constant)

- +ve paramagnets
- -ve diamagnets
- >> 0 ferromagnets

$$\vec{H} = \left(\frac{1}{\mu_0} \vec{B} - \vec{M} \right)$$

$$\vec{B} = \mu_0 (\vec{H} + \vec{M})$$

$$\vec{B} = \mu_0 (\vec{H} + \chi_m \vec{H})$$

$$\vec{B} = \mu_0 (1 + \chi_m) \vec{H} \rightarrow \boxed{\vec{B} = \mu \vec{H}} \rightarrow \boxed{\mu = \mu_0 (1 + \chi_m)} \quad \frac{\mu}{\mu_0} = (1 + \chi_m) = \mu_r$$

Permeability of the material

For vacuum: $\chi_m = 0$

35

Medium

$$\chi_m \neq \chi_m(\vec{r})$$

Homogeneous Medium

$$\chi_m = \chi_m(\vec{r})$$

Inhomogeneous Medium

$$\chi_m \neq \chi_m(\vec{H}, \vec{r}) \quad \vec{M} = \chi_m \vec{H}$$

Linear Medium

$$\vec{B} = \mu \vec{H}$$

paramagnets diamagnets

$$\chi_m = \chi_m(\vec{H}) \quad \vec{M} \neq \chi_m \vec{H}$$

Non-Linear Medium

$$\vec{B} = \mu \vec{H}$$

ferromagnets

$$\chi_m = \text{Constant} \quad \vec{M} \parallel \vec{H}$$

Isotropic Medium

$$\vec{B} \parallel \vec{H}$$

$$\chi_m = \text{tensor} \quad \vec{M} \not\parallel \vec{H}$$

Non-Isotropic Medium

$$\vec{B} \not\parallel \vec{H}$$

36

Boundary Condition

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$B_{\text{above}}^{\perp} = B_{\text{below}}^{\perp}$$

Continuous



$$\vec{B} = \mu \vec{H}$$

$$\mu_{\text{above}} H_{\text{above}}^{\perp} = \mu_{\text{below}} H_{\text{below}}^{\perp}$$

Discontinuous

$$\vec{\nabla} \times \vec{H} = \vec{J}_f$$

$$H_{\text{above}}^{\parallel} - H_{\text{below}}^{\parallel} = K_f$$

Discontinuous



$$\frac{B_{\text{above}}^{\parallel}}{\mu_{\text{above}}} - \frac{B_{\text{below}}^{\parallel}}{\mu_{\text{below}}} = K_f$$

Discontinuous

37

Boundary Condition

In the absence of any free current

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$B_{\text{above}}^{\perp} = B_{\text{below}}^{\perp}$$

Continuous



$$\vec{B} = \mu \vec{H}$$

$$\mu_{\text{above}} H_{\text{above}}^{\perp} = \mu_{\text{below}} H_{\text{below}}^{\perp}$$

Discontinuous

$$\vec{\nabla} \times \vec{H} = 0 \quad \vec{H} = -\vec{\nabla} V_{\text{mag}}$$

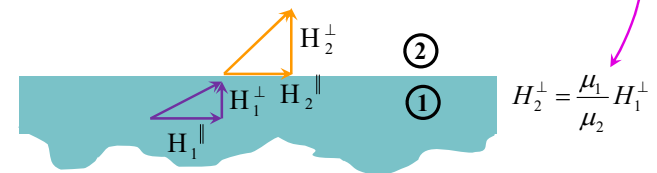
$$H_{\text{above}}^{\parallel} = H_{\text{below}}^{\parallel}$$

Continuous



$$\mu_{\text{below}} B_{\text{above}}^{\parallel} = \mu_{\text{above}} B_{\text{below}}^{\parallel}$$

Discontinuous



38

Boundary Condition

In the absence of any free current

- $B^{\perp}$ is always continuous across the boundary
- $H^{\perp}$ is discontinuous across the boundary
- $B^{\parallel}$ is discontinuous across the boundary
- $H^{\parallel}$ is always continuous across the boundary

- $D^{\perp}$ is always continuous across the boundary
- $E^{\perp}$ is discontinuous across the boundary
- $D^{\parallel}$ is discontinuous across the boundary
- $E^{\parallel}$ is always continuous across the boundary

$\vec{D}$ follows BC as was by $\vec{B}$ and $\vec{E}$ follows BC as was by $\vec{H}$

39

Boundary Condition

In the absence of any free charge

$$\vec{\nabla} \cdot \vec{D} = 0 \quad \vec{\nabla} \times \vec{E} = 0 \quad \epsilon_0 \vec{E} = \vec{D} - \vec{P}$$

In the absence of any free current

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \vec{\nabla} \times \vec{H} = 0 \quad \mu_0 \vec{H} = \vec{B} - \mu_0 \vec{M}$$

Replacing $\vec{D}$ to $\vec{B}$, $\vec{E}$ to $\vec{H}$, $\vec{P}$ to $\mu_0 \vec{M}$ and ϵ_0 to μ_0

Electrostatic Equations $\rightarrow$ Magnetostatic Equations

40

Boundary Condition

In the absence of any free current

This means, we have already
solved so many magnetostatics
problem

$$\vec{B} = \mu \vec{H}$$

$$\vec{D} = \epsilon \vec{E}$$

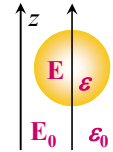
Replacing $\vec{D}$ to $\vec{B}$, $\vec{E}$ to $\vec{H}$, $\vec{P}$ to $\vec{\mu}_0 \vec{M}$ and ϵ_0 to μ_0
Electrostatic Equations $\rightarrow$ Magnetostatic Equations

41

Boundary Condition

Example: 1

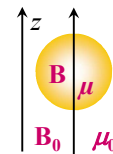
A sphere of homogeneous linear **dielectric material** is placed in an otherwise uniform field $\vec{E}_0$. Find the dipole moment of the sphere.



$$\vec{p} = \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \vec{E}_0 R^3$$

Example: 2

A sphere of homogeneous linear **magnetic material** is placed in an otherwise uniform field $\vec{B}_0$. Find the dipole moment of the sphere.



$$\vec{m} = \left(\frac{\mu - \mu_0}{\mu + 2\mu_0} \right) \vec{B}_0 R^3$$

42